4. Harmonic Solid and Phonon Modes

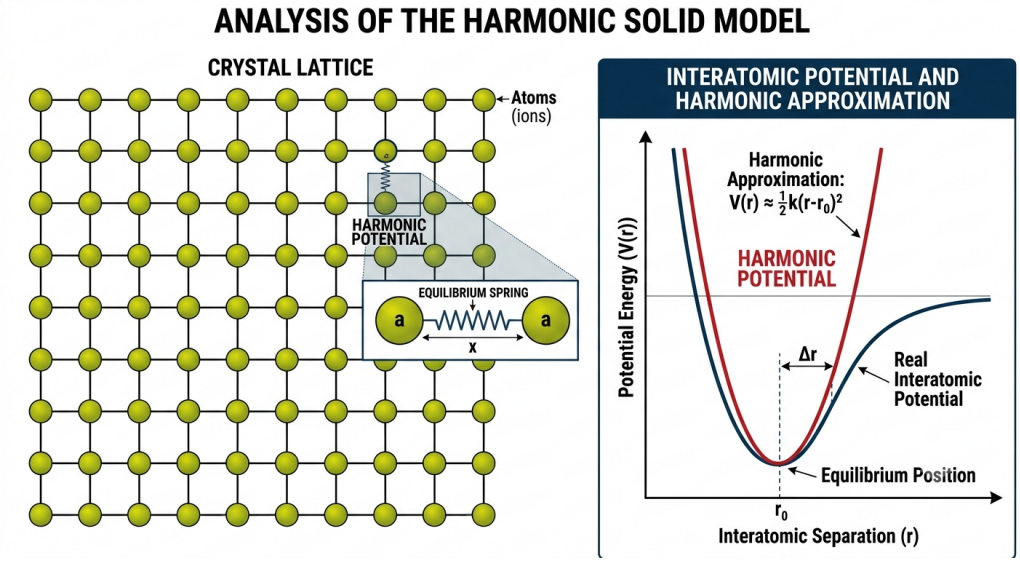


Figure 1: Schematic representation of a 1D harmonic solid showing atoms arranged in a periodic lattice with lattice spacing a, connected by interatomic springs representing the harmonic potential.

A. The Harmonic Approximation

In a crystalline lattice, atoms sit at equilibrium positions corresponding to potential energy minima at interatomic spacing $r = a$:

$$\left.\frac{dV(r)}{dr}\right|_{r=a} = 0$$

For small displacement fluctuations $u = r - a$, expanding $V(r)$ in a Taylor series about $u = 0$ yields:

$$V(u) \approx V(a) + \frac{1}{2}Cu^2 \quad \text{where} \quad C = \left.\frac{d^2V}{dr^2}\right|_{r=a}$$

Thus, the lattice potential is approximated by harmonic springs with force constant C.

B. Einstein's Model of Specific Heat

Einstein modeled a 3D solid of N atoms as $3N$ independent 1D quantum harmonic oscillators, all vibrating at an identical single frequency ω_E. The energy levels for each oscillator are:

$$\varepsilon_n = \left(n + \frac{1}{2}\right)\hbar\omega_E$$

The total internal energy U for $3N$ oscillators is given by:

$$U = 3N\left(\frac{1}{2}\hbar\omega_E + \frac{\hbar\omega_E}{e^{\hbar\omega_E/k_BT} - 1}\right) = 3Nk_B\Theta_E\left(\frac{1}{2} + \frac{1}{e^{\Theta_E/T} - 1}\right)$$

where $\Theta_E = \hbar\omega_E/k_B$ is the Einstein temperature. Differentiating with respect to T gives the heat capacity C_v:

$$C_v = \left(\frac{\partial U}{\partial T}\right)_V = 3Nk_B \left(\frac{\Theta_E}{T}\right)^2 \frac{e^{\Theta_E/T}}{\left(e^{\Theta_E/T}-1\right)^2}$$

C. High and Low Temperature Limits

- **High-Temperature Limit ($T \gg \Theta_E$):** Expanding $e^{\Theta_E/T} \approx 1 + \Theta_E/T$ yields:

  $$C_v \to 3Nk_B$$

  recovering the classical Dulong–Petit law.

- **Low-Temperature Limit ($T \ll \Theta_E$):** As $T \to 0$, $e^{\Theta_E/T} \gg 1$, giving:

  $$C_v \approx 3Nk_B \left(\frac{\Theta_E}{T}\right)^2 e^{-\Theta_E/T}$$

- **Failure at Low Temperatures:** The predicted exponential drop-off disagrees with experimental observations, which show $C_v \propto T^3$. This discrepancy arises because Einstein ignored coupled interatomic collective waves (long-wavelength acoustic phonons with $\omega \to 0$).

D. Debye Theory: 1D Monatomic Lattice and Phonons

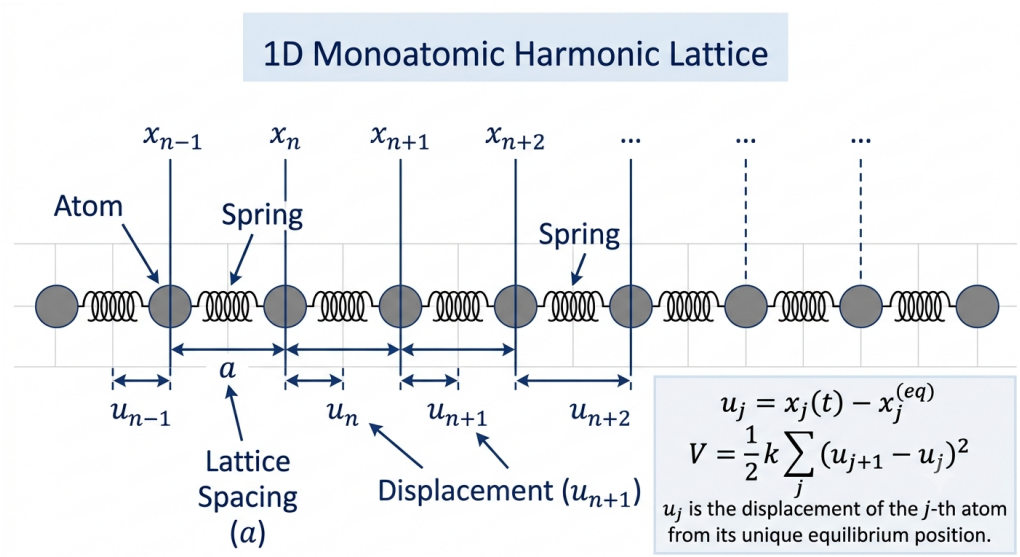


Figure 2: Schematic representation of a 1D monatomic chain with equilibrium spacing a, mass m, and spring constant $C = m\omega_0^2$. Displacements u_n measure deviations from equilibrium positions $x_n^{(0)} = na$.

E. Hamiltonian of the 1D Harmonic Lattice

Consider a 1D lattice of N identical atoms of mass m with equilibrium positions $x_n^{(0)} = na$. Let $u_n = x_n - na$ denote the displacement from equilibrium. The potential energy V with nearest-neighbor spring coupling ω_0 is:

$$V = \frac{1}{2} m\omega_0^2 \sum_n (u_{n+1} - u_n)^2$$

The classical Hamiltonian of the system is:

$$H = \sum_{n=0}^{N-1} \frac{p_n^2}{2m} + \frac{1}{2} m\omega_0^2 \sum_{n=0}^{N-1} (u_{n+1} - u_n)^2$$

F. Hamilton's Equations of Motion

From Hamilton's canonical equations:

$$\dot{u}_n = \frac{\partial H}{\partial p_n} = \frac{p_n}{m}$$

$$\dot{p}_n = -\frac{\partial H}{\partial u_n} = m\omega_0^2 (u_{n+1} + u_{n-1} - 2u_n)$$

Combining these gives the equation of motion:

$$m\ddot{u}_n = m\omega_0^2 (u_{n+1} + u_{n-1} - 2u_n)$$

G. Classical Phonon Dispersion Relation

Applying Born–von Karman periodic boundary conditions ($u_{n+N} = u_n$), we introduce Fourier normal modes:

$$u_n(t) = \frac{1}{\sqrt{N}} \sum_k U_k \, e^{i(kna - \omega t)}$$

Substituting this ansatz into the equation of motion yields:

$$-\omega^2 e^{ikna} = \omega_0^2 \left(e^{ik(n+1)a} + e^{ik(n-1)a} - 2e^{ikna} \right)$$

$$-\omega^2 = \omega_0^2 \left(e^{ika} + e^{-ika} - 2 \right) = -4\omega_0^2 \sin^2 \left(\frac{ka}{2} \right)$$

Taking the positive square root gives the acoustic dispersion relation:

$$\omega(k) = 2\omega_0 \left| \sin \left(\frac{ka}{2} \right) \right|$$

In the long-wavelength limit ($k \to 0$), $\omega(k) \approx (\omega_0 a)k = v_s k$, where v_s is the speed of sound.

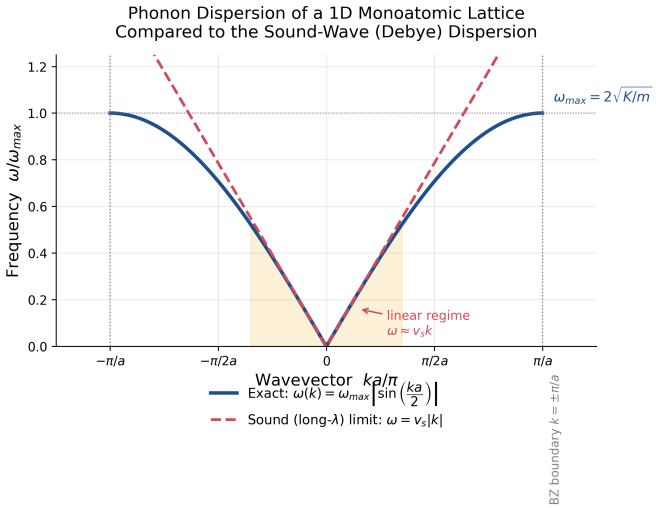


Figure 3: Phonon dispersion of 1D monoatomic lattice

H. Quantum theory of Phonons

1. Real-Space Canonical Quantum Operators

Consider a one-dimensional periodic chain consisting of N identical atoms, each of mass m, separated by an equilibrium lattice spacing a. We impose Born–von Karman periodic boundary conditions such that $\hat{u}_{n+N} = \hat{u}_n$ and $\hat{p}_{n+N} = \hat{p}_n$.

In quantum mechanics, the displacement operator $\hat{u}_n$ of the n-th atom relative to its equilibrium position $x_n^{(0)} = na$ and its canonical momentum operator $\hat{p}_n$ satisfy the equal-time canonical commutation relations:

$$[\hat{u}_n, \hat{p}_{n'}] = i\hbar\,\delta_{nn'}, \qquad [\hat{u}_n, \hat{u}_{n'}] = 0, \qquad [\hat{p}_n, \hat{p}_{n'}] = 0 \tag{1}$$

The total quantum mechanical Hamiltonian of the 1D lattice in real space under the harmonic approximation (nearest-neighbor spring constant $C = m\omega_0^2$) is given by:

$$\hat{H} = \sum_{n=0}^{N-1} \left[\frac{\hat{p}_n^2}{2m} + \frac{1}{2} m\omega_0^2 (\hat{u}_{n+1} - \hat{u}_n)^2 \right] \tag{2}$$

—

2. Reciprocal Space Fourier Transformation

To exploit translation symmetry, we expand the site-space operators $\hat{u}_n$ and $\hat{p}_n$ in terms of discrete Fourier normal modes in reciprocal space:

$$\hat{u}_n = \frac{1}{\sqrt{N}} \sum_k \hat{Q}_k\, e^{ikna} \tag{3}$$

$$\hat{p}_n = \frac{1}{\sqrt{N}} \sum_k \hat{P}_k\, e^{-ikna} \tag{4}$$

where k denotes the crystal wavevector restricted to the First Brillouin Zone:

$$k = \frac{2\pi p}{Na}, \qquad p \in \left(-\frac{N}{2}, \frac{N}{2} \right] \tag{5}$$

Hermiticity Constraints

Because the physical site operators $\hat{u}_n$ and $\hat{p}_n$ are Hermitian ($\hat{u}_n^\dagger = \hat{u}_n$ and $\hat{p}_n^\dagger = \hat{p}_n$), the modal operators in reciprocal space satisfy:

$$\hat{Q}_k^\dagger = \hat{Q}_{-k}, \qquad \hat{P}_k^\dagger = \hat{P}_{-k} \tag{6}$$

Reciprocal Commutation Relations

Using the lattice completeness/orthogonality identity:

$$\sum_{n=0}^{N-1} e^{i(k-k')na} = N\delta_{k,k'} \tag{7}$$

we evaluate the commutator between normal mode coordinate and momentum operators:

$$\begin{aligned}
[\hat{Q}_k, \hat{P}_{k'}] &= \left[\frac{1}{\sqrt{N}}\sum_{n=0}^{N-1} \hat{u}_n e^{-ikna},\ \frac{1}{\sqrt{N}}\sum_{n'=0}^{N-1} \hat{p}_{n'} e^{ik'n'a}\right] \\
&= \frac{1}{N}\sum_{n,n'} e^{-ikna} e^{ik'n'a} [\hat{u}_n, \hat{p}_{n'}] \\
&= \frac{i\hbar}{N}\sum_{n=0}^{N-1} e^{-i(k-k')na} = i\hbar\,\delta_{k,k'}
\end{aligned} \tag{8}$$

Similarly, $[\hat{Q}_k, \hat{Q}_{k'}] = 0$ and $[\hat{P}_k, \hat{P}_{k'}] = 0$.

—

3. Creation and Annihilation Operators

We define the ladder operators $\hat{a}_k$ (phonon annihilation operator) and $\hat{a}_k^\dagger$ (phonon creation operator) for each mode k:

$$\hat{a}_k = \sqrt{\frac{m\omega_k}{2\hbar}}\hat{Q}_k + i\frac{1}{\sqrt{2m\hbar\omega_k}}\hat{P}_{-k}^\dagger \tag{9}$$

$$\hat{a}_k^\dagger = \sqrt{\frac{m\omega_k}{2\hbar}}\hat{Q}_k^\dagger - i\frac{1}{\sqrt{2m\hbar\omega_k}}\hat{P}_{-k} \tag{10}$$

where ω_k is the characteristic frequency of mode k.

Bosonic Commutation Algebra

Using $[\hat{Q}_k, \hat{P}_{k'}] = i\hbar\delta_{k,k'}$, we verify the commutator $[\hat{a}_k, \hat{a}_{k'}^\dagger]$:

$$\begin{aligned}
[\hat{a}_k, \hat{a}_{k'}^\dagger] &= \left[\sqrt{\frac{m\omega_k}{2\hbar}}\hat{Q}_k + \frac{i}{\sqrt{2m\hbar\omega_k}}\hat{P}_k,\ \sqrt{\frac{m\omega_{k'}}{2\hbar}}\hat{Q}_{k'}^\dagger - \frac{i}{\sqrt{2m\hbar\omega_{k'}}}\hat{P}_{-k'}\right] \\
&= -\frac{i}{2\hbar}[\hat{Q}_k, \hat{P}_{k'}] + \frac{i}{2\hbar}[\hat{P}_k, \hat{Q}_{k'}^\dagger] \\
&= \frac{1}{2}\delta_{k,k'} + \frac{1}{2}\delta_{k,k'} = \delta_{k,k'}
\end{aligned} \tag{11}$$

Thus, the phonon operators satisfy canonical **bosonic commutation relations**:

$$[\hat{a}_k, \hat{a}_{k'}^\dagger] = \delta_{k,k'}, \qquad [\hat{a}_k, \hat{a}_{k'}] = 0, \qquad [\hat{a}_k^\dagger, \hat{a}_{k'}^\dagger] = 0 \tag{12}$$

—

4. Expansion of Field Operators

Inverting the definitions of $\hat{a}_k$ and $\hat{a}_k^\dagger$ expresses the reciprocal operators as:

$$\hat{Q}_k = \sqrt{\frac{\hbar}{2m\omega_k}} \left(\hat{a}_k + \hat{a}_{-k}^\dagger \right) \tag{13}$$

$$\hat{P}_k = -i\sqrt{\frac{m\hbar\omega_k}{2}} \left(\hat{a}_{-k} - \hat{a}_k^\dagger \right) \tag{14}$$

Substituting these back into the real-space Fourier transforms yields the second-quantized representations of displacement and momentum:

$$\hat{u}_n = \sum_k \sqrt{\frac{\hbar}{2mN\omega_k}} \left(\hat{a}_k e^{ikna} + \hat{a}_k^\dagger e^{-ikna} \right) \tag{15}$$

$$\hat{p}_n = -i \sum_k \sqrt{\frac{m\hbar\omega_k}{2N}} \left(\hat{a}_k e^{ikna} - \hat{a}_k^\dagger e^{-ikna} \right) \tag{16}$$

—

5. Diagonalization of the Lattice Hamiltonian

We now substitute $\hat{u}_n$ and $\hat{p}_n$ into the Hamiltonian $\hat{H} = \hat{T} + \hat{V}$.

A. Kinetic Energy ($\hat{T}$)

$$\begin{aligned} \hat{T} &= \sum_{n=0}^{N-1} \frac{\hat{p}_n^2}{2m} \\ &= \frac{1}{2m} \sum_{n=0}^{N-1} \left(-i \sum_k \sqrt{\frac{m\hbar\omega_k}{2N}} (\hat{a}_k e^{ikna} - \hat{a}_k^\dagger e^{-ikna}) \right) \left(-i \sum_{k'} \sqrt{\frac{m\hbar\omega_{k'}}{2N}} (\hat{a}_{k'} e^{ik'na} - \hat{a}_{k'}^\dagger e^{-ik'na}) \right) \end{aligned} \tag{17}$$

Using the orthogonality relation $\sum_n e^{i(k+k')na} = N\delta_{k,-k'}$, the sum simplifies to:

$$\hat{T} = \sum_k \frac{\hbar\omega_k}{4} \left(\hat{a}_k \hat{a}_{-k} + \hat{a}_k^\dagger \hat{a}_{-k}^\dagger + \hat{a}_k \hat{a}_k^\dagger + \hat{a}_k^\dagger \hat{a}_k \right) \tag{18}$$

B. Potential Energy ($\hat{V}$)

The displacement difference operator is:

$$\hat{u}_{n+1} - \hat{u}_n = \sum_k \sqrt{\frac{\hbar}{2mN\omega_k}} (e^{ika} - 1) \left(\hat{a}_k e^{ikna} - \hat{a}_{-k}^\dagger e^{ikna} \right) \tag{19}$$

Noting that $|e^{ika} - 1|^2 = 2 - 2\cos(ka) = 4\sin^2\left(\frac{ka}{2}\right)$, the potential energy sums to:

$$\hat{V} = \sum_k \frac{m\omega_0^2\hbar}{2m\omega_k} \left[4\sin^2\left(\frac{ka}{2}\right) \right] \frac{1}{4} \left(-\hat{a}_k \hat{a}_{-k} - \hat{a}_k^\dagger \hat{a}_{-k}^\dagger + \hat{a}_k \hat{a}_k^\dagger + \hat{a}_k^\dagger \hat{a}_k \right) \tag{20}$$

C. Dispersion Enforcing and Final Diagonalization

To ensure complete cancellation of the non-diagonal creation/annihilation product terms ($\hat{a}_k\hat{a}_{-k}$ and $\hat{a}_k^\dagger\hat{a}_{-k}^\dagger$), we define the dispersion relation as:

$$\omega_k^2 = 4\omega_0^2 \sin^2\left(\frac{ka}{2}\right) \implies \omega_k = 2\omega_0 \left|\sin\left(\frac{ka}{2}\right)\right| \tag{21}$$

With this dispersion relation, the off-diagonal terms cancel identically, leaving:

$$\hat{H} = \sum_k \frac{\hbar\omega_k}{2}\left(\hat{a}_k^\dagger\hat{a}_k + \hat{a}_k\hat{a}_k^\dagger\right) \tag{22}$$

Applying $[\hat{a}_k, \hat{a}_k^\dagger] = 1 \implies \hat{a}_k\hat{a}_k^\dagger = \hat{a}_k^\dagger\hat{a}_k + 1$, we obtain the diagonalized second-quantized Hamiltonian:

$$\hat{H} = \sum_k \hbar\omega_k \left(\hat{a}_k^\dagger\hat{a}_k + \frac{1}{2}\right) \tag{23}$$

—

6. Summary and Physical Interpretation

1. **Quasiparticle Picture:** The coupled system of N interacting classical atomic harmonic oscillators is exactly mapped into a system of N uncoupled quantum harmonic oscillators in reciprocal space.

2. **Phonon Number Operator:** The term $\hat{n}_k = \hat{a}_k^\dagger\hat{a}_k$ is the number operator whose eigenvalues $n_k \in \{0, 1, 2, \dots\}$ denote the number of phonons present in mode k.

3. **Zero-Point Energy:** Even in the ground state ($n_k = 0$), the lattice possesses a finite quantum zero-point energy:

$$E_0 = \sum_k \frac{1}{2}\hbar\omega_k \tag{24}$$

4. **Phonon Dispersion:** A single quantum of vibrational excitation carrying crystal momentum $\hbar k$ has an energy $\hbar\omega_k$, where $\omega_k = 2\omega_0|\sin(ka/2)|$.